

Algebra 1
Unit 1
Lesson 7 Practice

Name **Answer Key** _____
Date _____
Period _____

1. Write the expression $27^{\frac{2}{3}}$ in two different equivalent forms.

Expression 1	Expression 2
$\sqrt[3]{27^2}$	$\left(27^{\frac{1}{3}}\right)^2$

2. The value of $27^{\frac{2}{3}}$ is

- ☐ 9
- ☐ 32
- ☐ 54
- ☐ 72

For questions 3-5, write an equivalent radical expression and then find the value of the expression.

3. $(-125)^{\frac{2}{3}} = \sqrt[3]{-125^2} = 25$

4. $-\left(\frac{3,125}{32}\right)^{\frac{2}{5}} = \sqrt[5]{-\left(\frac{3,125}{32}\right)^2} = -\frac{25}{4}$

5. $(81)^{-\frac{3}{2}} = \frac{1}{\sqrt{81^3}} = \frac{1}{729}$

Evaluate each expression in questions 6-10 without a calculator.

$$6. \quad (4^{-2}) (49^{-\frac{1}{2}}) = \frac{1}{4^2} \cdot 49^{-\frac{1}{2}} = \frac{1}{16} \cdot \frac{1}{\sqrt{49}} = \frac{1}{16} \cdot \frac{1}{7} = \frac{1}{112}$$

$$7. \quad \left(256^{\frac{1}{4}}\right)^3 \cdot 6^{-1} = 256^{\frac{3}{4}} \cdot \frac{1}{6} = \sqrt[4]{256^3} \cdot \frac{1}{6} = \sqrt[4]{(4 \cdot 4 \cdot 4 \cdot 4)^3} \cdot \frac{1}{6} = 4^3 \cdot \frac{1}{6} = 64 \cdot \frac{1}{6} = \frac{64}{6} = 10\frac{4}{6} = 10\frac{2}{3} \text{ or } \frac{32}{3}$$

$$8. \quad -\left(\frac{1}{4}\right)^{\frac{7}{2}} = \frac{-1^{\frac{-7}{2}}}{4^{\frac{7}{2}}} = -4^{\frac{7}{2}} = -\sqrt{4^7} = -2^7 = -128$$

$$9. \quad (-1^{15}) \cdot \left(-64^{\frac{1}{3}}\right) = -1 \cdot \sqrt[3]{-64^1} = -1 \cdot \sqrt[3]{-4 \cdot -4 \cdot -4} = -1 \cdot -4 = 4$$

$$10. \quad \left(\frac{5}{35}\right)^{-2} \cdot (2^{-2}) = \frac{5^{-2}}{35^{-2}} \cdot 2^{-2} = \frac{35^2}{5^2} \cdot \frac{1}{2^2} = \frac{1225}{25} \cdot \frac{1}{4} = \frac{1225}{100} = 12.25 \text{ or } \frac{49}{4}$$